

Element AD

$\Pi_e(u_e)$

exact local gradient and Hessian

higher-order elements can be costly

Constitutive AD

$\psi(\varepsilon_q)$

$B_q^\top C_q B_q$ assembly

lowest-cost route in fixed 3D endpoint comparison

Colored SFD

rank-local probe HVPs only where needed

parallel coloring / recovery

useful when exact Hessians are too expensive